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Hexadecimal system: base 16

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
0 1 2 3 4 5 6 7 8 9 A B C D E F

Integers

We can represent an integer as $\pm S_{k-1} \dots S_1 S_0$. The value is calculated as:

$$N = \pm S_{k-1} \times 16^{k-1} + S_{k-2} \times 16^{k-2} + \dots + S_2 \times 16^2 + S_1 \times 16^1 + S_0 \times 16^0$$

Maximum value

The maximum value of a hexadecimal integer with k digits is $N_{\max} = 16^k - 1$. For example, if $k = 5$, then the maximum value is $N_{\max} = 16^5 - 1 = 1048575$.

Octal system: base 8 set of symbols is $S = \{0, 1, 2, 3, 4, 5, 6, 7\}$

Integers

We can represent an integer as $\pm S_{k-1} \dots S_1 S_0$. The value is calculated as:

$$N = \pm S_{k-1} \times 8^{k-1} + S_{k-2} \times 8^{k-2} + \dots + S_2 \times 8^2 + S_1 \times 8^1 + S_0 \times 8^0$$

Maximum value

The maximum value of an octal integer with k digits is $N_{\max} = 8^k - 1$. For example, if $k = 5$, then the maximum value is $N_{\max} = 8^5 - 1 = 32767$.

Comparison:

Table 2.2 Comparison of numbers in the four systems

Decimal	Binary	Octal	Hexadecimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4

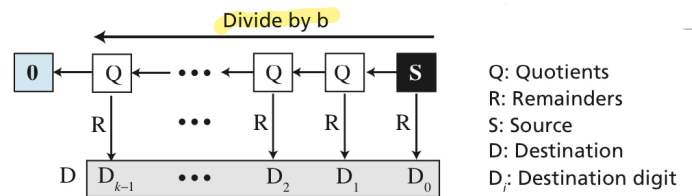
2.2 Positional Number Sys

Table 2.2 Comparison of numbers in the four systems (Continued)

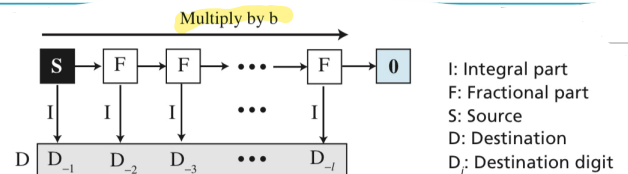
Decimal	Binary	Octal	Hexadecimal
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Gr

any base to decimal: $\sum_{k=1} s_k \times b^k + \dots + \sum_0 s_0 \times b^0 + s_{-1} \times b^{-1} \dots s_{-k} \times b^{-k}$
 decimal to any base: Integer:



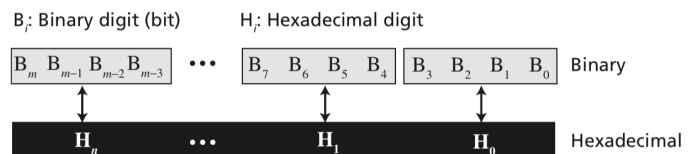
Fraction:



Note:
The fraction may never become zero.
Stop when enough digits have been created.

binary - hexadecimal

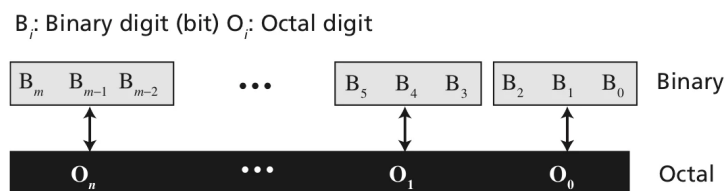
Figure 2.10 Binary to hexadecimal and hexadecimal to binary



Binary-octal conversion

We can easily convert a number from binary to octal and *vice versa*. The reason is that there is an interesting relationship between the two bases: three bits is one octal digit. Figure 2.11 shows how this conversion can be done.

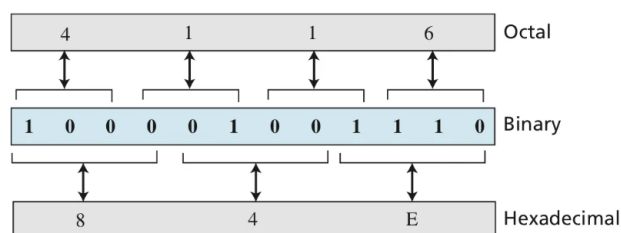
Figure 2.11 Binary to hexadecimal conversion



Octal-hexadecimal conversion

It is not difficult to convert a number in octal to hexadecimal or *vice versa*. We can use the binary system as the intermediate system. Figure 2.12 shows an example.

Figure 2.12 Octal to hexadecimal and hexadecimal to octal conversion



The following illustrates the process:

- ❑ To convert from octal to hexadecimal, we first convert the number in the octal system to binary. We then rearrange the bits in groups of four bits to find the hexadecimal equivalent.
- ❑ To convert from hexadecimal to octal, we first convert the number in the hexadecimal system to binary. We then rearrange the bits in groups of three to find the octal equivalent.

Number of digits : We can find the number of digits of base b by using relation $k = \lceil \log_b N \rceil$. N is decimal symbol of integer ex. $\lceil \log_{10} 234 \rceil = \lceil 2.37 \rceil = 3$

$$\lceil \log_2 234 \rceil = \lceil 7.8 \rceil = 8 \Rightarrow (11101010)_2$$

Number of digits for conversion : If we convert one base to another, we need to know the maximum number of digits of source system and minimum number of digits of destination system, then:

$$\begin{cases} \text{maximum: } b_1^k - 1 \\ \text{minimum: } b_2^x - 1 \end{cases} \quad \text{then } b_2^x - 1 \geq b_1^k - 1$$

$$\Rightarrow x \geq k \cdot \left(\frac{\log b_1}{\log b_2} \right) / x = \lceil k \cdot \left(\frac{\log b_1}{\log b_2} \right) \rceil$$

* We have to know the minimum number of digits we need to avoid overflow while converting.